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Game Design Document

<https://www.gamedesigning.org/learn/game-design-document/>

Questions to Ask yourself (throughout the game development process)

- Who is it made for
 - Is it designed for the audience it's made for
- Is it fun
- What's the players motivation/goal
- Is it accessible

Title

- "Sime Research"

Concept Statement

What is your concept? Is it a puzzle game in which an alien must find his way home? Or is it a game like Outlast in which the main character needs to find crucial information in an 'abandoned' hospital? This is where you outline what the main idea of the game will be.

- Fantasy Slime-based sokoban game
- Organizing slime
- Educational Goal: Learn the science concept behind the combo.
- Player Goal: Solve the level by creating the right chain of combinations
- Early levels: simple 2-element combos (Steam, Mud, Lava).
- Mid levels: introduce disasters and natural forces (Tornado, Hurricane, Volcano).
- Later levels: multi-element combos as the "boss puzzles" (Rainbow, Tsunami, Forging, Ecosystem).
- Each unlock = cool science fact + visual
- "Help me (I need to) move the slimes to the correct spot. When they are in the correct spot they'll start to react to each other. And something cool will happen"
- "Each slime represents a different element: the dark blue one is water, the dark green one is earth, the light blue one is air/wind, and the red one is fire."
- "Just in case you forget I'll leave a key on the side for you."
- "Will you help me with my slime research?"
- "Hi, my name is Mage. And I'm a student at the Magic Academy. I need to complete my research in order to graduate. Will you help me?"
- Use WASD to move around. Hold R and move your mouse to rotate the screen. Hold M and move the mouse to move the screen.

Genre

- Puzzle, 2D, Education, Alchemy

Mechanics

The game mechanics refer to different aspects of the game. Very simply, they are the visuals, sounds, animations, physics, and more. In specific detail.

- Look at “slime ranger sokoban” for visual example
- Maybe have colored dots represents which elements that were combined (above the combined spot)
- The player click on already combined slime and the undo button becomes highlighted and the slime go back to where they started
- Slightly up and down movement from slime (animation)
- Little dust trail when the player moves (animation)
- Fantasy stuff in the background (visual)

The Story

Delving deeper, you need to flesh out the descriptions of the story, setting, and who the characters will eventually be. What will your story be? Will it have multiple character arcs, or just one?

- Help the mage with their slime research. They are researching what happens when they push certain types of slime together. They already have some slime in different locations. They just need help pushing the slime without somewhere to go and pushing them together.

Characters:

characters all have distinct personalities, and you must do the same for your characters. What are their wants, needs, and motivations?

This is where you fully flesh out our protagonists, antagonists, ancillary characters, and more. This is where your roster of loveable, (or perhaps not so loveable), characters will go to grow and develop into finished products.

- **Main Character: Mage**
 - Wants to discover what happens when you combine certain slime together
 - Needs help moving the slime to the correct research spot
 - Motivated/excited to work with slime and see what happens
 - Slime researcher (Scientist/Academic)
 - Needs help with researcher
 - Experiments with slime to see what the reaction is when you combine the slime
- **NPC: Slime** (4 different type)
 - Fire (redish/orange)
 - Water (blue)
 - Earth (green)
 - Air (super light blue/grey)

Setting:

- Based in a magical fantasy world. Within that world the player is in a fantasy type forest with trees, tree stumps, and bushes.

Target Audience:

- Puzzle & STEM fans (ages 12+) — a casual but smart game that's both fun and lightly educational. Middle school age 11-14 Science

Levels:

- Blue = water (water clusters)
- Red = fire (carbon combustion)
- Green = earth (mineral structures)
- Light Blue = air (gas clusters, ozone)
- Used this site to sketch out levels: <https://www.sokobanonline.com/build>

TUTORIAL

- 1. "Hi there! I'm Mage, a student at the Magic Academy. I need to complete my research, and I appreciate you lending a hand."
- 2. "I'm currently studying four variants: water, air, fire, and earth."
- 3. "For each experiment, I want to observe the reaction produced when two or more slimes are combined. As we go I'll also pose a few questions that are part of the research."
- 4. "To move around, use WASD."
- 5. "You'll need to push the slimes onto their matching colored spots. Careful though — you can only push one slime at a time, and you can't pull them back."
- 6. "There's a key in the top right corner that shows you what each slime is, in case you forget."
- 7. "If you ever want to take a break or jump to a different level, just hit the pause button in the top left corner."
- 8. "Okay. I think that's everything. Why don't you give it a try?"

EASY (Water Clusters)

1. 4 water molecules

- "What do you think will happen if we combine 4 water molecules together?"
 - A. They form a cyclic tetramer — a stable, ring-shaped cluster held together by hydrogen bonds
 - B. They form a covalent bond and become a new compound
 - C. They immediately crystallize into ice, regardless of temperature
 - D. They split apart into hydrogen gas and oxygen gas
- (Correct: A)
 - "It's actually more stable than smaller clusters."
 - "Fun fact, if you're curious — scientists use lasers and spectroscopy to study water clusters this small!"

2. 2 water molecules

- "What do you suppose happens when just 2 water molecules bond together?"
 - A. They form a dimer — the smallest possible hydrogen-bonded water cluster
 - B. They form a single molecule of hydrogen peroxide

- C. They repel each other due to opposite charges
- D. They can only bond below freezing temperature
- (Correct: A)
 - "It's fascinating, how every cloud, raindrop, and glass of water is made of countless dimers, constantly forming and breaking apart."

3. 3 water molecules

- "What do you think happens when 3 water molecules bond together?"
 - A. They form the smallest stable ring structure a water cluster can make
 - B. They form a linear, straight-line chain instead of a ring
 - C. They become more reactive and unstable than a single water molecule
 - D. They form a compound with a different chemical formula
- (Correct: A)
 - "Though I should mention — these tiny rings only last a fraction of a second before breaking apart and reforming. Blink and you'll miss it."

4. 5 water molecules

- "And what do you think happens once we're up to 5 water molecules?"
 - A. The cluster starts branching into a more complex, 3D shape
 - B. It automatically forms a hexagonal ice crystal
 - C. It becomes less stable than a 2- or 3-molecule cluster
 - D. The molecules stop hydrogen bonding at this size
- (Correct: A)
 - "It's actually the beginning of something rather beautiful — this branching is part of what eventually leads to the six-sided symmetry of real snowflakes."

5. 4 water molecules (bond strength)

- "How do you think hydrogen bond strength changes when 4 water molecules cluster together, compared to just 2?"
 - A. The bonds get stronger through cooperative hydrogen bonding, where each molecule reinforces its neighbors
 - B. The bonds weaken because there are more molecules competing for space
 - C. Bond strength stays exactly the same no matter the cluster size
 - D. The molecules switch from hydrogen bonds to covalent bonds
- (Correct: A)
 - "I like to think of it as a bit of a team effect — it's why larger water clusters are actually harder to pull apart than smaller ones."

MEDIUM (Element Combinations)

1. 5 fire + 4 earth

- "What do you think happens when carbon — our fire slime — combines with mineral material, the earth slime?"

- A. Carbon can become chemically trapped in minerals, similar to how carbon sequesters into carbonate rock like limestone
- B. The carbon instantly converts the mineral into a metal
- C. The mixture becomes radioactive
- D. The carbon burns off completely, leaving no trace
- (Correct: A)
 - "It's a rather important process, actually. Over long timescales, it helps remove carbon dioxide from the atmosphere."

2. 2 air + 2 water + 2 earth + 2 fire

- "Now, what do you suppose happens when air, water, earth, and fire all interact together, in balanced amounts?"
 - A. They drive natural processes like weathering and erosion that gradually break rock down into soil
 - B. They immediately neutralize each other and stop reacting
 - C. They combine into a single new chemical element
 - D. The reaction is always explosive and destructive
- (Correct: A)
 - "Gradually is the key word, mind you. It can take hundreds, even thousands of years to form just one inch of soil."

3. 2 fire + 2 earth

- "What do you think happens when heat is applied to mineral material?"
 - A. The material can melt and later cool into igneous rock, similar to lava solidifying
 - B. It instantly forms a pure diamond structure
 - C. It becomes an inert gas with no further reactivity
 - D. It dissolves completely, even without water present
- (Correct: A)
 - "Obsidian is a lovely example, if you're curious — it's natural volcanic glass, formed when lava cools so quickly the crystals never get the chance to form."

4. 4 earth + 4 water

- "What do you think happens when a larger amount of mineral material meets water?"
 - A. Minerals break down and settle out, similar to erosion and sediment formation
 - B. The water evaporates instantly regardless of temperature
 - C. It forms a crystalline diamond lattice
 - D. The minerals convert directly into gas
- (Correct: A)

- "The Grand Canyon, believe it or not, was carved almost entirely this way — by water-driven erosion, over millions of years."

5. 3 earth + 3 water

- "And on a smaller scale — what do you think happens with less earth and water together?"
 - A. A small-scale sediment or mud forms — an early stage of soil or clay development
 - B. It triggers a sudden flash flood
 - C. The minerals harden instantly into diamond
 - D. The water freezes permanently regardless of temperature
- (Correct: A)
 - "Clay, in fact, is made from some of the very smallest mineral particles this process produces."

HARD (Multi-Element Systems)

1. 3 earth + 2 fire + 2 air + 3 water

- "This one's a bit more involved. What do you think happens when earth, fire, air, and water all interact together, in these proportions?"
 - A. They form the basic conditions needed for a small ecosystem — supporting soil formation and early life
 - B. They instantly cancel out and leave no trace
 - C. They fuse into a single solid metal
 - D. The combination is inherently unstable and explosive
- (Correct: A)
 - "It's thought that a similar mix of conditions is what helped make early Earth habitable for life in the first place."

2. 2 earth + 2 water + 2 air

- "What do you think happens when earth, water, and air interact in equal amounts?"
 - A. Basic soil begins forming through mechanical and chemical weathering
 - B. It produces a sandstorm
 - C. The materials fuse into glass
 - D. The materials dissolve into nothing, leaving no byproduct
- (Correct: A)
 - "Plant roots can actually speed this along, by physically cracking the rock apart."

3. 2 fire + 2 earth + 2 air + 2 water

- "What do you suppose happens when fire, earth, air, and water interact in equal amounts?"
 - A. This combination mirrors real volcanic activity, where magma interacts with groundwater and atmosphere

- B. The elements neutralize into an inert, unreactive gas
- C. They form a diamond
- D. The mixture freezes solid regardless of temperature
- (Correct: A)
 - "When magma meets groundwater, it can actually trigger explosive eruptions — we call those phreatomagmatic eruptions."

4. 3 earth + 3 water + 1 air

- "What do you think happens when earth and water dominate, with only a touch of air present?"
 - A. A low-oxygen, muddy environment forms, similar to a wetland or swamp ecosystem
 - B. It triggers a drought
 - C. The water evaporates instantly despite low air presence
 - D. It solidifies directly into solid rock
- (Correct: A)
 - "Wetlands are remarkable, really — they act like natural sponges, soaking up floodwater and protecting the communities nearby."

5. 5 earth + 5 fire

- "What do you think happens when a large amount of mineral material meets intense heat?"
 - A. The rock melts and reforms — the basic process behind magma formation and igneous rock
 - B. It triggers an earthquake
 - C. It instantly forms a diamond
 - D. The earth material extinguishes the fire/heat source
- (Correct: A)
 - "Some igneous rock cools so quickly, it forms natural glass instead of crystals. Rather elegant, if you ask me."

SPECIAL (Classical Elements — All Four Equal)

1. 1 fire + 1 air + 1 earth + 1 water (origins)

- "Here's a bit of history for you — what did ancient philosophers believe about these four elements?"
 - A. They believed all matter in the universe was composed of these four basic elements
 - B. They believed these four elements would always destroy each other on contact
 - C. They believed only one of the four was "real" and the others were illusions
 - D. They believed these elements only existed on other planets
- (Correct: A)

- "It's a rather old idea — over two thousand years, tracing back to ancient Greek philosophy."

2. 1 fire + 1 air + 1 earth + 1 water (Earth systems)

- "What do you think these four elements correspond to, in real Earth science?"
 - A. Earth's major systems: atmosphere (air), hydrosphere (water), lithosphere (earth), and energy from the sun (fire)
 - B. Four unrelated substances with no scientific connection
 - C. A single classification of igneous rock
 - D. A weather pattern exclusive to desert biomes
- (Correct: A)
 - "Scientists still use this very model today, to study climate and weather."

3. 1 fire + 1 air + 1 earth + 1 water (storms)

- "What real weather phenomenon do you think depends on all four working together?"
 - A. A thunderstorm — requiring heat energy, moisture, atmosphere, and temperature differences near the land
 - B. An earthquake
 - C. A solar eclipse
 - D. A landslide
- (Correct: A)
 - "A single lightning bolt, incidentally, can be hotter than the surface of the sun."

4. 1 fire + 1 air + 1 earth + 1 water (cycles)

- "Which named scientific cycle do you think depends on all four elements interacting?"
 - A. The water cycle — evaporation (fire/heat), atmospheric movement (air), precipitation onto land (earth), and water itself
 - B. The rock cycle exclusively
 - C. The carbon cycle exclusively
 - D. None of these elements interact in any named cycle
- (Correct: A)
 - "Rather humbling, when you think about it — the water you drink today could be the very same water dinosaurs once drank."

5. 1 fire + 1 air + 1 earth + 1 water (balance)

- "What do you think the result is when all four elements exist in a stable, balanced state?"
 - A. A self-sustaining ecosystem capable of supporting life
 - B. Total ecological collapse
 - C. Permanent separation of all four elements
 - D. No measurable effect on the environment

- (Correct: A)
 - "As far as we know, Earth is the only place where this exact balance comes together. Rather special, when you think of it that way."

User Interface:

This is the display that gamers will use and see during their playtime. These are menus, health bars, advice panels, tutorials, and more. An easy to use user interface will score major points with gamers, trust me.

- During Gameplay
 - Display minimum number of moves that are possible to make (maybe only show when they completed the level)
 - Display number of moves made
 - Back arrow (For going back one move)
 - Forward arrow (For replaying the move they went back on)
 - Restart arrow circle (for starting from the beginning of that level)
 - The slime that can be moved will either blink or have steal light on them
 - Key on the side saying what element each slime is
- Before Gameplay
 - Tutorial/Instructions Visual on how to play
 - Play button
 - Chooses if they want to play the easy, medium or hard levels
 - When one is chosen it list out all the levels available (visual screenshot of each level)
 - Saves the information on that browser
 - Might need to add save button
 - Credits
- After Gameplay
 - Shows the moves made and the possible minimum moves
 - Display/shows the results from combining the slime
 - Continue button
 - Home button
 - Replay button

Technical:

These are the aspects of game design like which game engine to use, what programming language to utilize, and more. This is where the coders shine. Will it use JavaScript? C++?

- Unity Engine
- Pixelorama
- Blender
- Game assets from website (Particle Effects and Audio)

Assets:

These are actually the assets used in-game. For example, assets can be specific objects and in-game items.

- No in game items/assets

Project Status Report

Think of this as a regular progress report. You can even add some milestones for you and the team to complete to get an idea or roadmap of the project better.

- Create/find images (slime, grass, tree, tree stump, bush, ect.)
 - Upload image
 - Place images
- Make levels
- Code
- Find audio
 - Upload audio
 - Place audio

TO-DO List

- Art
 - Slime
 - Ground (Unity Asset Store Texture)
 - Sky/Skybox (Unity Asset Store Texture)
 - Grass blades(I made grass blades)
 - Trees
 - Bushes
 - Rocks
 - Mage/Wizard (player)
 - Menu image
 - Pixel-ish speech box (Scroll or old paper) and font
 - Pixel-ish visual example of answer
 - Sticker/badge
 - Particle Effects
- Code
 - Player movement (can move and only push no pull)
 - Slime movement (move only when player pushes it not pull or anything else)
 - Slime Goal Placement (the spot will only activate when in the correct spot/same color)
 - Game management (interaction between player and slime)
 - Audio (controls audio)
 - Particle effects (Look for)
- Audio
 - Gameplay music
 - Menu, credits, instructions/controls music
 - Button click sound effect
 - Push sound effect
 - Walk sound effect

- Correct goal sound effect
- Code first
 - Make one level first
 - Once it works, input everything (Art, Audio, etc.)
 - After that copy and paste the level and adjust to make the other levels
 - Use shapes as place holders (Use different colors)
 - Cubes (Walls)
 - Cylinders (Player)
 - Sphere or different shape (Slime) trigger to be darker instead of a different color
 - Use a grid for visual
 - Make levels
 - Menu
 - Make something like the level select for sticker/badge
 - Key
 - Blue = water (water clusters)
 - Red = fire (carbon combustion)
 - Green = earth (mineral structures)
 - Light Blue = air (gas clusters, ozone)
 - Credits
 - Audio
 - Explosion: ejah_music (Pixabay Website)
 - Walk: joentnt (Pixabay Website)
 - Push: OxidVideos (Pixabay Website)
 - Goal: Universfield (Pixabay Website)
 - Button: freesoundeffects (Pixabay Website)
 - Gameplay Music: Sascha Ende (Ende Website)
 - Menu Music: The_Mountain (Pixabay Website)
 - Skybox: Jermesa Studio (Unity Assets Store)
 - Ground: Ink Phantom (Unity Assets Store)
 - Particle Effect: Giant Grey (Unity Assets Store)
 - Speech Bubbles: DreamNoms (Unity Assets Store)
 - Font: Polahtype (Font Space Website)
 - instruction/tutorial
 - Rotate Plane (something extra to add left and right on screen buttons to rotate the camera so its a square and one click rotates the camera to one side)